ABSTRACT

Introducing the real-time web search engine, a groundbreaking solution revolutionizing information retrieval in the digital age. Traditional search engines, while effective, often struggle to keep pace with the rapidly evolving landscape of online content due to indexing delays and static ranking algorithms. Our real-time web search engine offers a dynamic approach, continuously fetching and updating web content as it becomes available, ensuring users receive the most up-to-date information in response to their queries. Leveraging advanced crawling and indexing techniques, the engine prioritizes freshness and relevance, delivering timely results that reflect the ever-changing nature of the web. Moreover, dynamic ranking algorithms factor in recency, user engagement metrics, and contextual relevance to provide personalized and contextually relevant search results. With features like personalized search experiences, real-time notifications for trending topics, and a responsive user interface, our real-time web search engine empowers users to stay informed and discover relevant content as it emerges, reshaping the way people interact with online information in today's fast-paced digital landscape.

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CHAPTER-1

INTRODUCTION

In an age where information serves as the lifeblood of our digital society, the demand for real-time access to relevant and accurate data has never been more pressing. Conventional web search engines, while foundational, struggle to keep pace with the lightning-fast evolution of online content, often resulting in outdated or irrelevant search results. Recognizing this challenge, we proudly introduce the Real-time Web Search Engine, a cutting-edge solution poised to redefine the landscape of information retrieval.

At its core, the Real-time Web Search Engine represents a paradigm shift in how we interact with and consume online information. By leveraging state-of-the-art technologies such as continuous crawling, dynamic indexing, and adaptive ranking algorithms, this innovative platform ensures that users receive the most up-to-date and contextually relevant search results instantaneously.

Unlike traditional search engines that rely on periodic updates and static ranking algorithms, the Real-time Web Search Engine operates in real-time, constantly scouring the web for fresh content and dynamically adjusting search rankings based on recency, relevance, and user engagement metrics. This dynamic approach not only ensures the freshness of search results but also enhances the overall search experience by delivering personalized and contextually relevant information tailored to the user's preferences and browsing history.

Moreover, the Real-time Web Search Engine goes beyond mere search functionality, offering a suite of features designed to empower users in their quest for knowledge and information. From real-time notifications for trending topics and breaking news to personalized search experiences and responsive user interfaces, this platform is engineered to meet the evolving needs and expectations of today's digital-savvy users.

In this paper, we delve into the intricate architecture, functionality, and underlying principles of the Real-time Web Search Engine, exploring how it leverages cutting-edge technologies to deliver unparalleled speed, accuracy, and relevance in information retrieval. We also examine the potential implications and applications of this revolutionary platform across various domains, from journalism and research to e-commerce and social media.

How It Works:

Crawling:

Web search engines use "web crawlers" or "spiders" to browse the internet and gather information from web pages. These crawlers follow links from one page to another, collecting data about the content, structure, and other metadata.

Indexing:

Once data is gathered, it's processed and stored in a database known as the "index." The index is optimized for fast retrieval, allowing the search engine to quickly find relevant pages when a query is made.

Query Processing:

When a user enters a search query, the search engine parses and interprets the input to understand what the user is looking for. This may involve breaking down the query into individual keywords, analyzing the context, and identifying the type of information requested.

Ranking and Retrieval:

The search engine uses various algorithms to determine which results are most relevant to the user's query. This involves factors like keyword matching, page relevance, popularity, user behavior, and more. The results are then sorted and presented to the user in order of relevance.

Display of Results:

The search engine shows the retrieved results in a format that's easy to understand. This typically includes a list of web pages with titles, snippets, and URLs, along with other features like images, videos, or news articles if relevant.

User Input:

User input for a search engine typically consists of a search query, which can be a single word, a phrase, or a question. The user might also use advanced search operators or filters to refine their search.

Classification: Classification in a search engine refers to categorizing and organizing content based on various criteria. This can involve classifying web pages by topic, type, language, or relevance to a specific search query.

1.1 What is Python

These are some facts about Python. Python is present most widely used multi-purpose and high-level programming language.it allows programming in Object-Oriented and Procedural paradigms. Python programs commonly are smaller than other programming languages like Java. Programmers have to write relatively less and indentation requirement of the language, makes them readable all the time. Python language is being used by the almost all tech-giant companies like – Google, Amazon, Facebook, Instagram, Dropbox, Uber… etc. The biggest strength of Python language is huge collection of standard libraries it can be used for the following

- Machine Learning.
- GUI Applications (such as Kivy, Tkinter, PyQt, etc.)
- Web frame works such as Django (used by YouTube, Instagram, Dropbox)
- Image processing (such as Open cv and Pillow)
- Web scraping (such as Scrapy, Beautiful So up and Selenium)
- Test Framework

1.1.1 Advantages of Python

Let us see how Python out performs other languages**.**

Advantages of Python Over Other Languages

1) Less Coding

When performing the same task in other languages, all tasks performed in Python require less coding. Python is also excellent standard library support, so there is no need to search for third-party libraries to get the job done. This is the main reason why many people recommend that beginners learn Python.

2) Affordable

Python is a free resource, so individuals, small businesses, or large organizations can use available free resources to build applications, and Python is a popular and widely used community supply.

3) Python is for Everyone

Python can run in any environment, be it Linux, Mac or Windows. Programmers need to learn different languages for different jobs, but with Python, you can professionally use Python to create web applications, perform data analysis and machine learning, automate things, perform web scraping, and create powerful games and visualization initial later rain programming language.

1.1.2 Disadvantages of Python

We have seen why Python is the best choice for our project. But if you allocate it, you should also be aware of its consequences. Let us now look at the disadvantages of choosing.

Disadvantages of Python Over Other Language

1. Speed Limitations

We have seen that Python code executes line by line. But because Python is interpreted, its execution speed is very slow. This is not a problem, unless speed is the focus of the project work. In other words, the speed is necessary, and the benefits Python provides are enough to distract us from its speed limitations.

2. Weak in Mobile Computing and Browsers

Although Python is a good server-side language, Python is rarely seen on the client side. Most importantly, is rarely used to implement smartphone- based applications. One of these applications is called Carbon nelle. Although Bright on exists, it is not so famous because it is not so safe.

3. Design Restrictions

Python is a dynamically typed language. This means that you do not need to declare variable types when writing code content. It uses to duck typing. what is that? Well, it just means that if it looks like a duck, then it must be a duck. This is easy for programmers in the coding process, and it will generate runtime errors.

4. Under developed data base access layer

Compared to the most widely used technologies such as JDBC (Java DataBase Connectivity) and ODBC (Open DataBase Connectivity), Python's database access layer is a bit underdeveloped and it is used less in large companies.

5. Simple

We are not kidding. Python's simplicity is indeed a problem. I don't do Java, I'm more of a Python person. To me, its syntax is so simple that Java code seems unnecessary.

1.1.3 Install Python on Windows and Mac Step by Step:

Step-1: Go to the official site and use Google Chrome or any other web browser to download and install Python. Or click on the following link:

https://www.python.org/

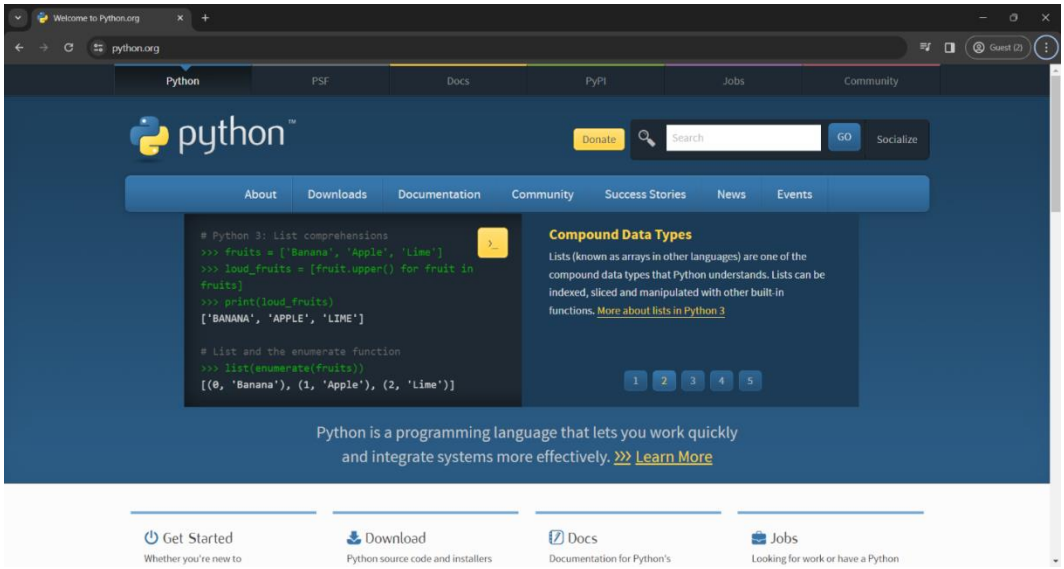


Fig 0.1 Open Python website to download the python latest version

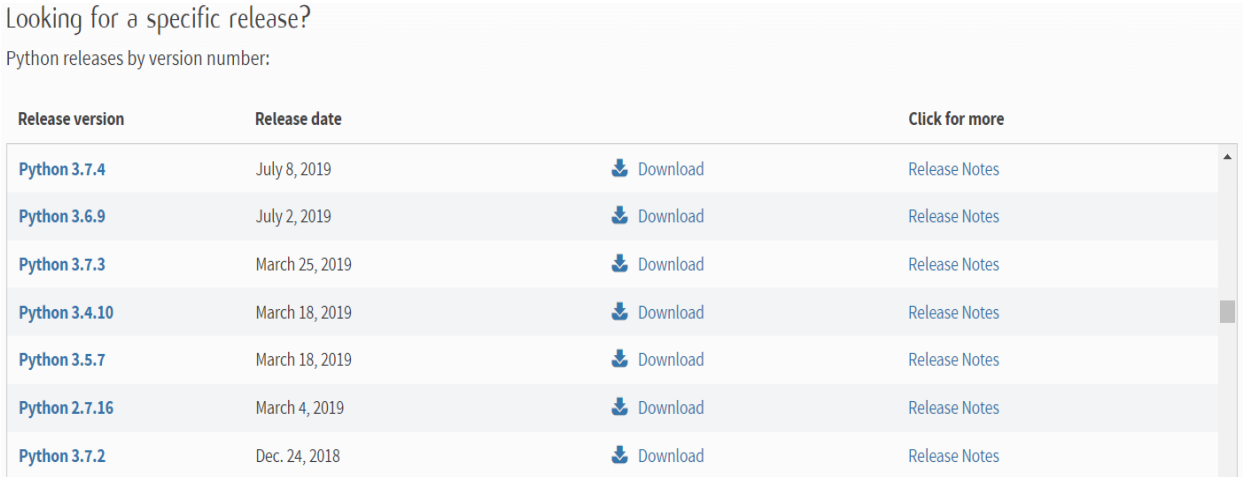
Now check the latest and correct version of your operating system.

Step-2: Click on the Download Tab.



Fig 0.2 Click on Download Python

Step-3: You can select the yellow Download Python for Windows3.7.4 button, or you can scroll download click the download of the corresponding version. Here, we are downloading the latest version of Python for Window 3.7.4.



Looking for a specific release?

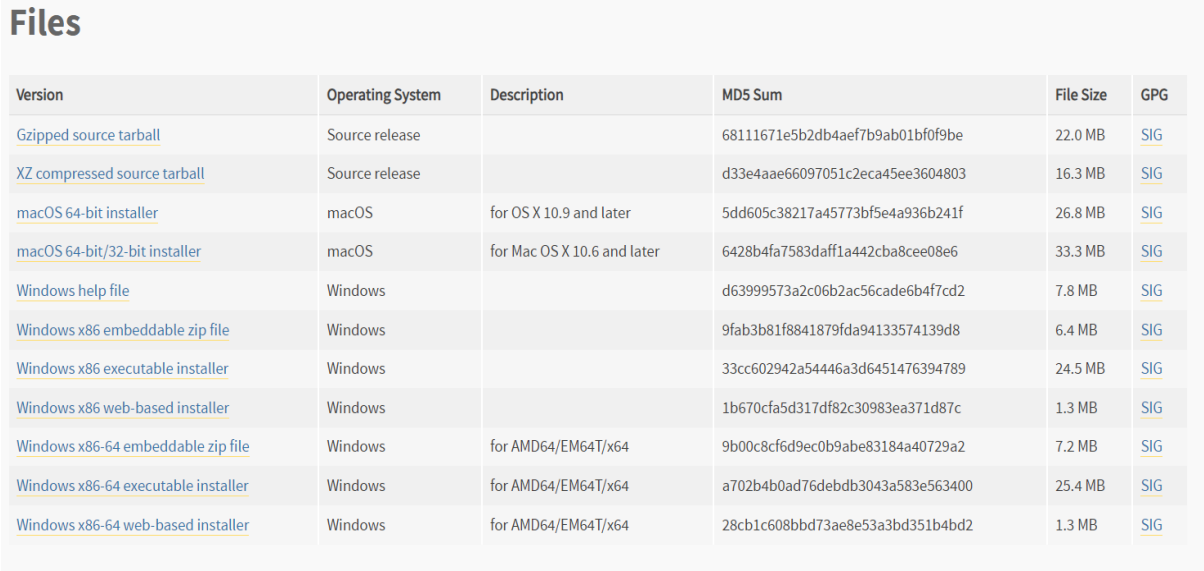
Python releases by version number:

Release version	Release date		Click for more
Python 3.7.4	July 8, 2019	Download	Release Notes
Python 3.6.9	July 2, 2019	Download	Release Notes
Python 3.7.3	March 25, 2019	Download	Release Notes
Python 3.4.10	March 18, 2019	Download	Release Notes
Python 3.5.7	March 18, 2019	Download	Release Notes
Python 2.7.16	March 4, 2019	Download	Release Notes
Python 3.7.2	Dec. 24, 2018	Download	Release Notes

Fig 0.3 Select the version what you want

Step-4: Scroll down the page until you find the "File" option.

Step5: Here you will see different versions of Python and operating systems.



Files

Version	Operating System	Description	MD5 Sum	File Size	GPG
Gzipped source tarball	Source release		68111671e5b2db4aef7b9ab01bf0f9be	22.0 MB	SIG
XZ compressed source tarball	Source release		d33e4aae66097051c2eca45ee3604803	16.3 MB	SIG
macOS 64-bit installer	macOS	for OS X 10.9 and later	5dd605c38217a45773bf5e4a936b241f	26.8 MB	SIG
macOS 64-bit/32-bit installer	macOS	for Mac OS X 10.6 and later	6428b4fa7583daff1a442cba8cee08e6	33.3 MB	SIG
Windows help file	Windows		d63999573a2c06b2ac56cade6b4f7cd2	7.8 MB	SIG
Windows x86 embeddable zip file	Windows		9fab3b81f8841879fda94133574139d8	6.4 MB	SIG
Windows x86 executable installer	Windows		33cc602942a54446a3d6451476394789	24.5 MB	SIG
Windows x86 web-based installer	Windows		1b670cfa5d317df82c30983ea371d87c	1.3 MB	SIG
Windows x86-64 embeddable zip file	Windows	for AMD64/EM64T/x64	9b00c8cf6d9ec0b9abe83184a40729a2	7.2 MB	SIG
Windows x86-64 executable installer	Windows	for AMD64/EM64T/x64	a702b4b0ad76debdb3043a583e563400	25.4 MB	SIG
Windows x86-64 web-based installer	Windows	for AMD64/EM64T/x64	28cb1c608bbd73ae8e53a3bd351b4bd2	1.3 MB	SIG

Fig 0.4 Python Versions

- To download Windows 32bit Python, you can select the built-in Windows X86 Zip file, WindowsX86 executable installer or WindowsX86 installer on the Web.
- TodownloadWindows64 bit Python, you can select any option of the three options. Zip File Embeddable Windows X866, Windows X8664 executable installer or

Windows X8664 installer based on the web.

Here you will install the installer based on the Windows X8664 website. Here, the first part of the Python version was completed. Now we will go in advance the second part when installing PythonI. Note: You can click on the option of the release of the version to know the changes or updates made in the version.

1.1.4 Installation of Python

Step-1: Goto Download and Open the downloaded python version to carry out the installation process.

Step-2: Before you click on Install Now, make sure to put a tick on Add Python 3.7 to Path

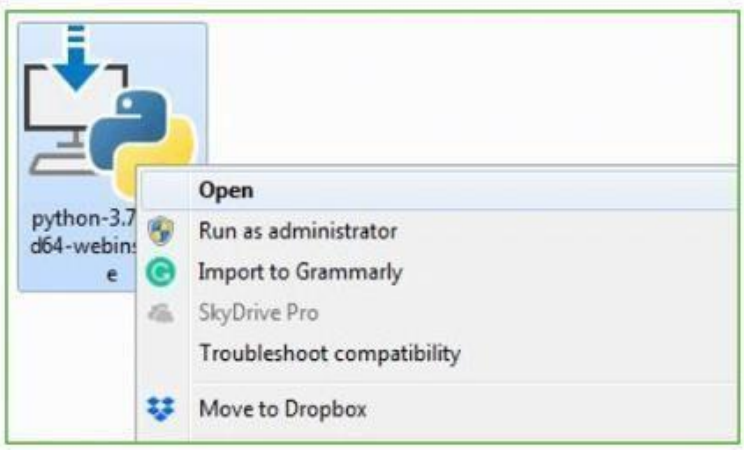


Fig 0.5 Click on ok to install Python



Fig 0.6 Click on Install Now to install

Step-3: Click on Install Now After the installation is successful. Click on Close.



Fig 0.7 After installation successful. Click on Close

With these above three steps on python installation, you have successfully and correctly installed Python. Now is the time to verify the installation.

Note: The installation process might take a couple of minutes.

1.1.5 Verify the Python Installation

Step-1: Click on Start

Step-2: In the Windows Run Command, type "cmd".

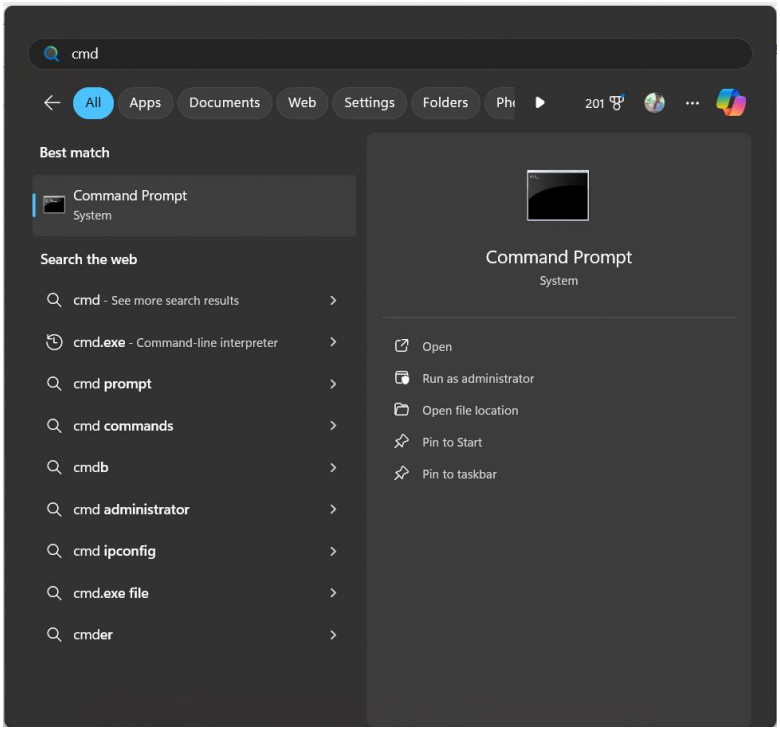


Fig 0.8 Open Command Prompt

Step-3: Open the Command prompt option.

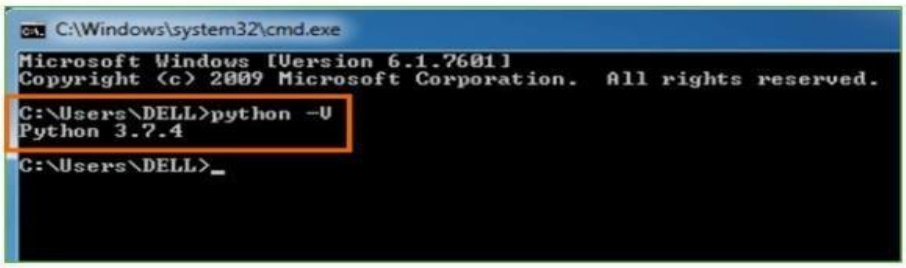


Fig 0.9 Type Python -V to check the python is correctly installed or not

Step-4: Let us test whether the python is correctly installed. Type **python– V** and press Enter.

Step-5: You will get the answer as 3.7.4

Note: If you have any of the earlier versions of Python already installed. You must first uninstall the earlier version and then install the new one.

1.1.6 Check how the Python IDLE works

Step-1: Click on Start

Step-2: In the Windows Run command, type "python idle".



Fig 0.10 Click on IDLE (Python3.7 64-bit) and launch the program

Step-3: Click on IDLE (Python3.7 64-bit) and launch the program

Step-4: To go ahead with working in IDLE you must first save the file. Click on File Click on Save

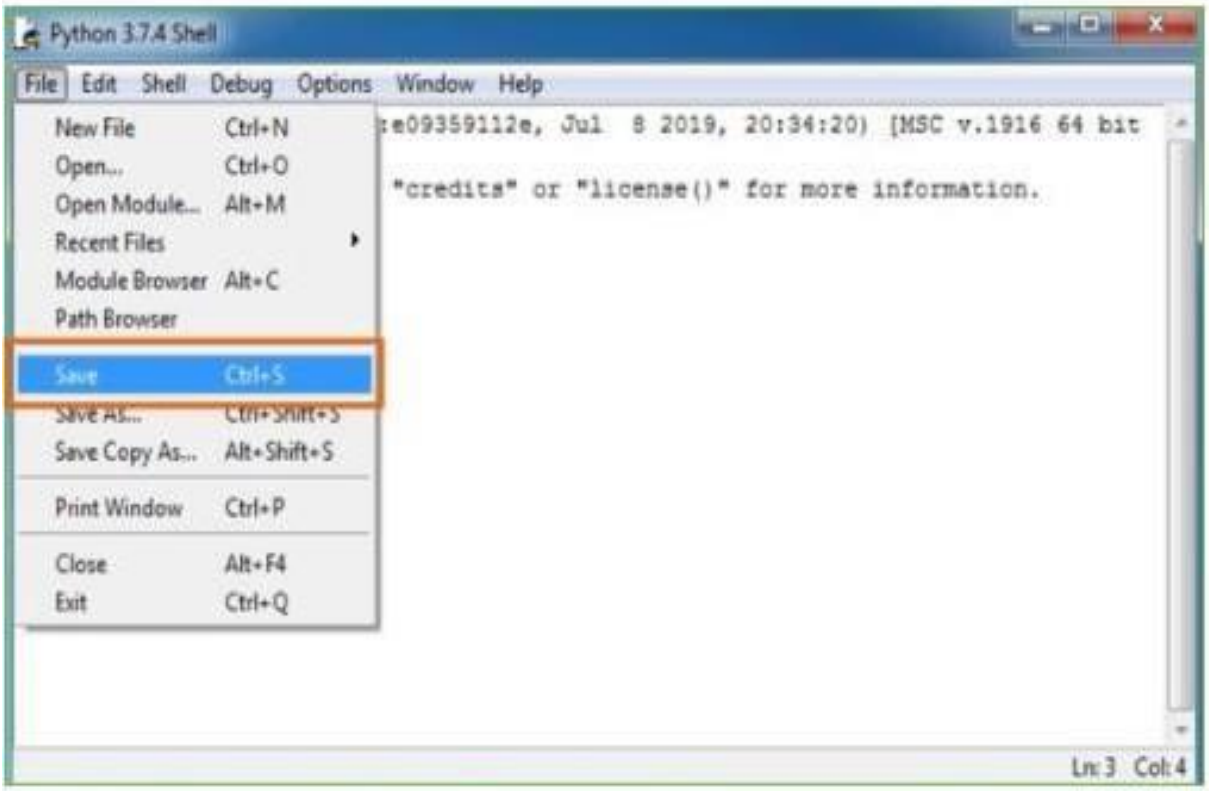


Fig 0.11 Click on File and Save the file

Step-5: Name the file and save as type should be Python files.

Click on SAVE. Here I have named the files as Hey World.

1.2 What is VS Code:

Visual Studio Code is the most popular code editor and the IDEs provided by Microsoft for writing different programs and languages. It allows the users to develop new code bases for their applications and allow them to successfully optimize them.

For its high popularity, individuals opt to Install Visual Studio Code on Windows over any other IDE. Installation of Windows Visual Studio Code is not a difficult matter. The Installation process starts with Downloading the Visual Studio Code EXE file to some on-screen instructions.

We are going to list all the steps required to Install VS Code on Windows in a detailed format.

- Quick Preview to Install Visual Studio Code on Windows
- Download the VS Code file from the Official Website.
- Execute the download file.
- Accept the Terms & Conditions.
- Click on the Install button.
- Wait for the installation to complete.
- Click on the Launch button to start it.

1.2.1 Steps to Install Visual Studio Code on Windows:

Step 1: Visit the Official Website of the Visual Studio Code using any web browser like Google Chrome, Microsoft Edge, etc.

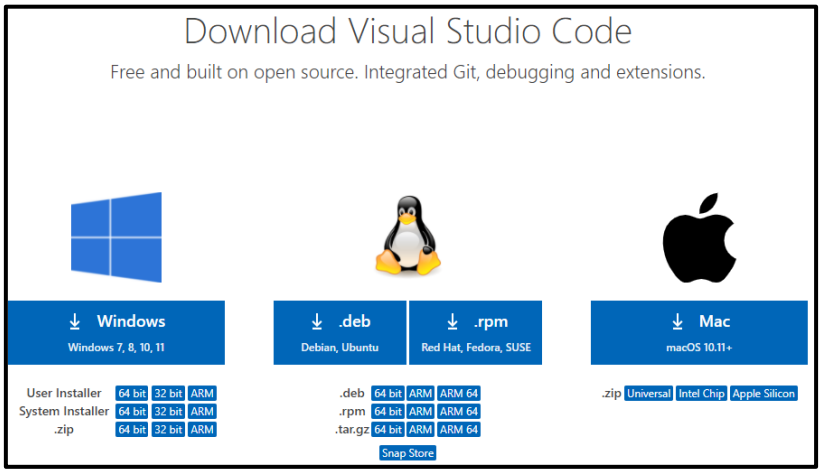


Fig 0.12 Visit the Official Website to download the VS Code

Step 2: Press the "Download for Windows" button on the website to start the download of the Visual Studio Code Application.

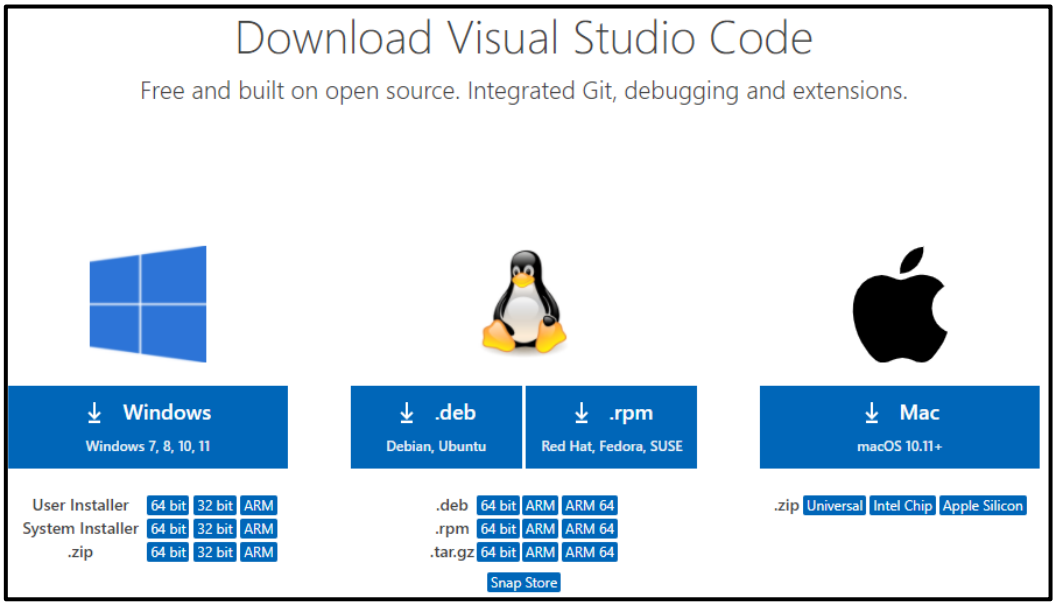


Fig 0.13 Click on Download for Windows

Step 3: When the download finishes, then the Visual Studio Code Icon appears in the downloads folder.



Fig 0.14 Click on VS Code Icon to start installation process

Step 4: Click on the Installer icon to start the installation process of the Visual Studio Code.

Step 5: After the Installer opens, it will ask you to accept the terms and conditions of the Visual Studio Code. Click on I accept the agreement and then click the Next button.

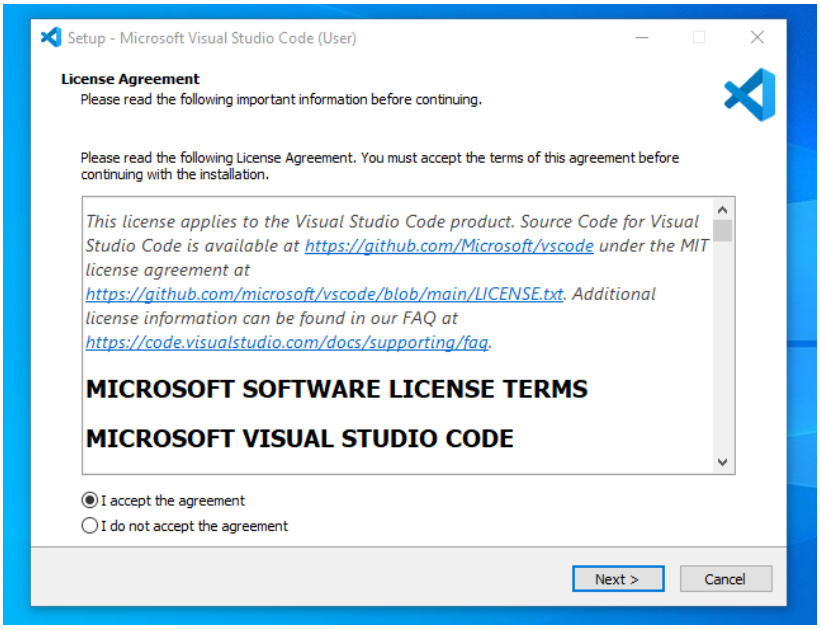


Fig 0.15 Accept the Agreement and click on Next

Step 6: Choose the location data for running the Visual Studio Code. It will then ask you to browse the location. Then click on the Next button.

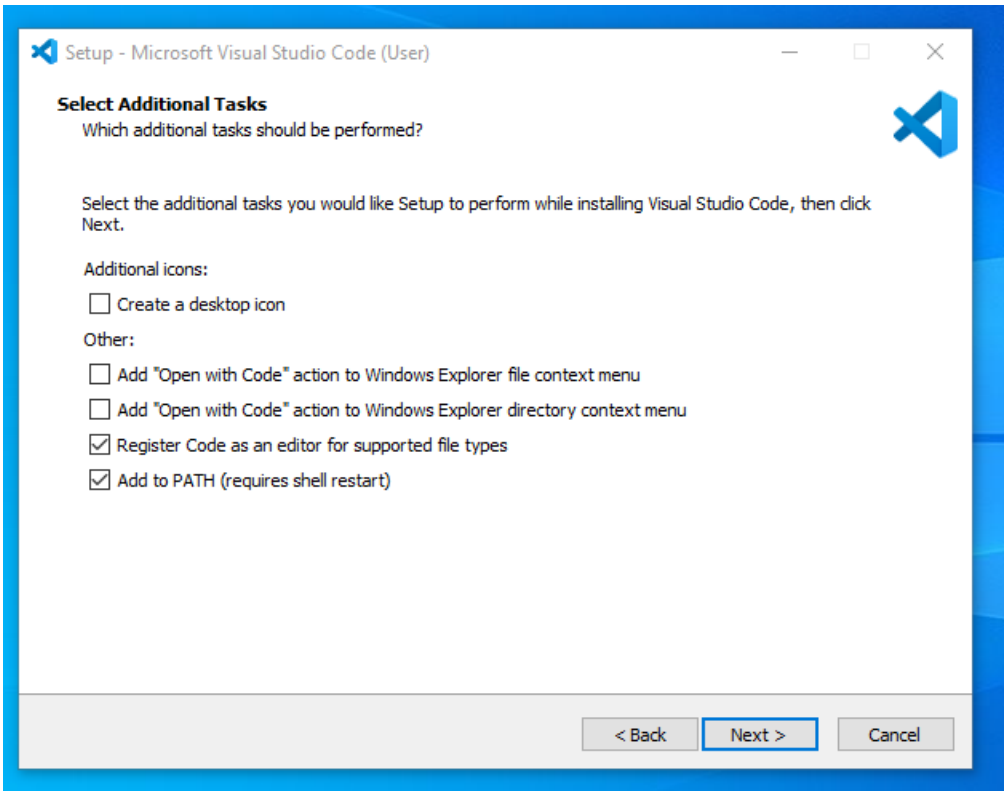


Fig 0.16 Choose the location and then Click on Next

Step 7: Then it will ask to begin the installation setup. Click on the Install button.

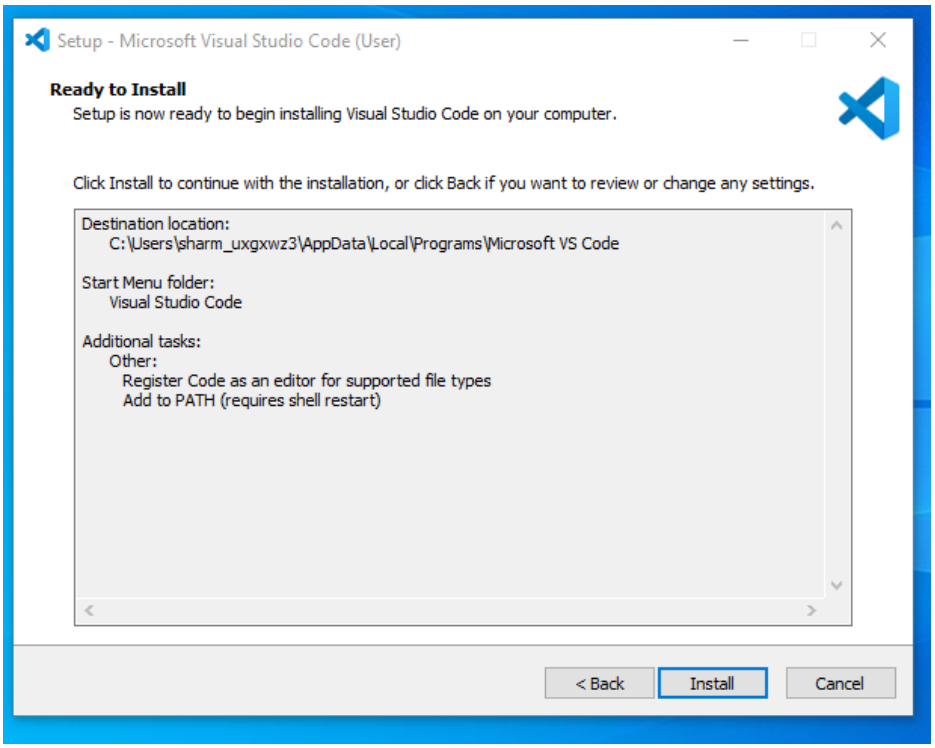


Fig 0.17 Click on Install

Step 8: After clicking on Install, it will take about 1 minute to install the Visual Studio Code on your device.

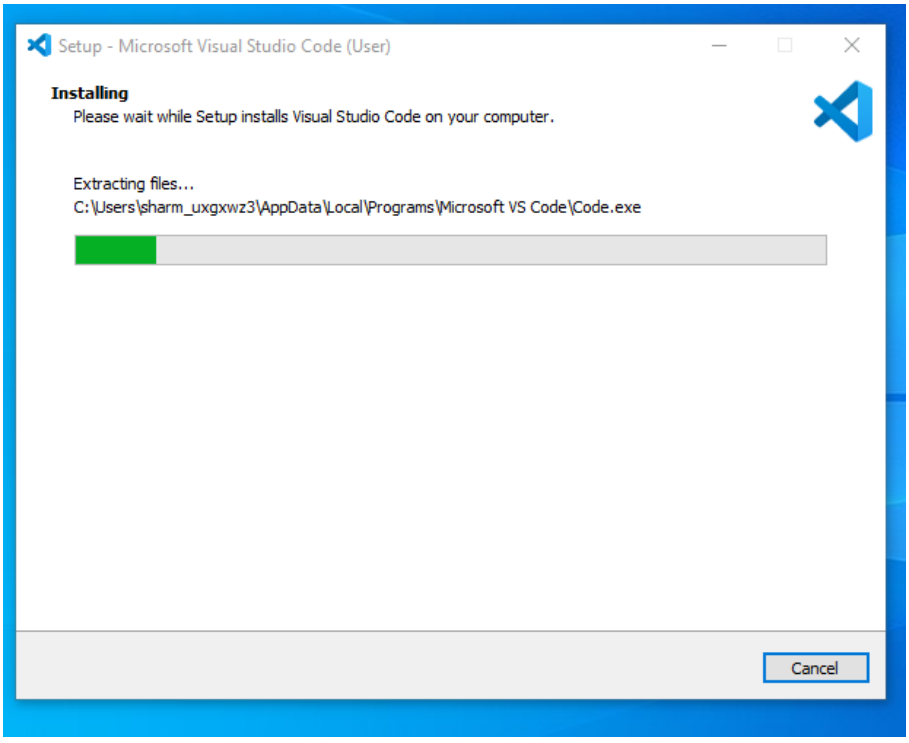


Fig 0.18 Installing VS Code

Step 9: After the Installation setup for Visual Studio Code is finished, it will show a window like this below. Tick the "Launch Visual Studio Code" checkbox and then click Next.

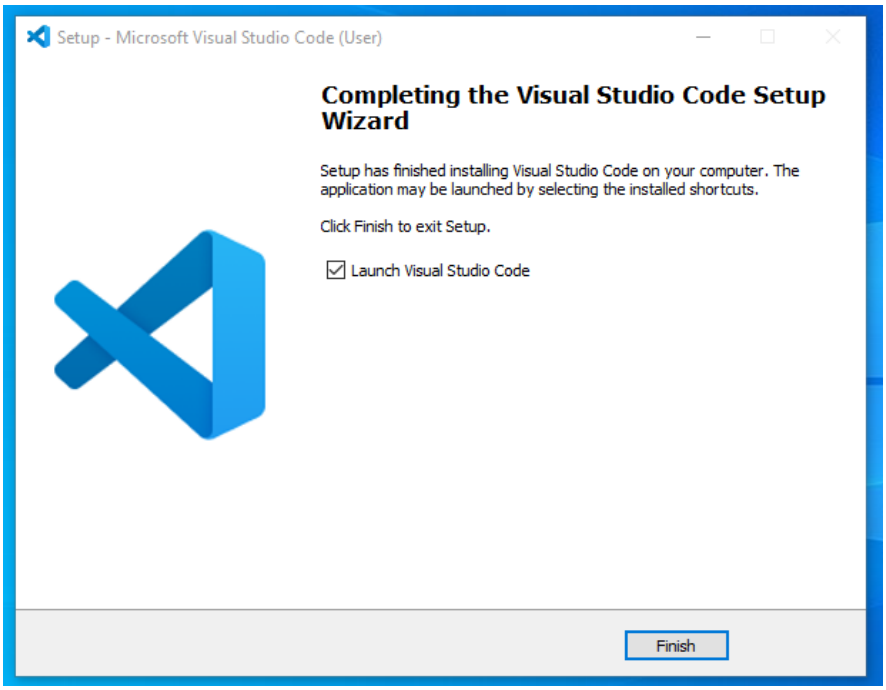


Fig 0.19 After Installation if Finished Click on Finish

Step 10: After the previous step, the Visual Studio Code window opens successfully. Now you can create a new file in the Visual Studio Code window and choose a language of yours to begin your programming journey!

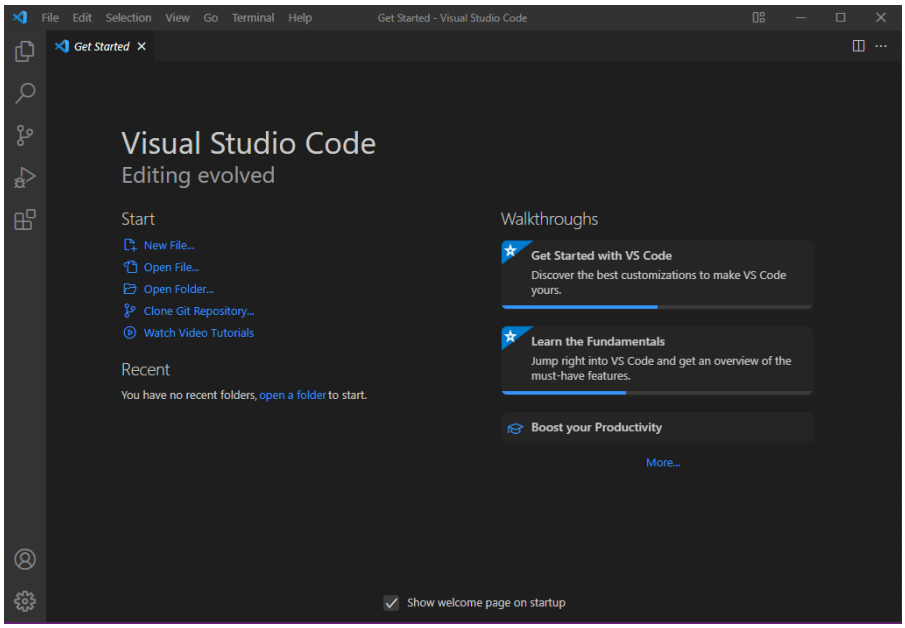


Fig 0.20 Visual Studio Code Screen

CHAPTER-2
LITERATURE SURVEY

2.1 Effective tool for exploring web: An Evaluation of Search engines Learning

(Dr. Sabha Ali and Ubaid Ullah Shah)

Search engines can be evaluated with the help of some methods which are very essential for the process of evaluation and therefore, Azimzadeh, Badie and Esnaashari (2016) discuss that search engines can be evaluated with the help of some automatic methods viz; "methods based on user's feedbacks" (which is very effective if a search engine which is to be evaluated is being used by a number of people on everyday basis), "methods based on consensus" (which is effective only if a search engine under evaluation performs better in comparison to all other search engines, "methods based on rank aggregation" (this method is used for those search engines which have high level of overlaps among results) and "methods based on known item searches" (this method can be used in case of navigational queries because methods which use navigational queries are more accurate and easy to execute).

In view of Xie (2004) interface design, system performance and collection coverage are the important components for users that help them to evaluate online information retrieval systems. Vaughan (2004) presents two newly methods viz; "the quality of result ranking" and "the ability to retrieve top ranked pages" for evaluating the retrieval performance of search engines and found that these two measures are more effective in evaluating web search engines and are capable to differentiate search engine performances.

Lopez-Pellicer, Florczyk, Bejar, Muro-Medrano and Zarazaga-Soria (2011) have performed an automated evaluation of three search engines viz; Google, Yahoo and Bing using their application programming interfaces and reveal that the discovery of geographic web services in search engines does not require the use of advanced search operators. While evaluating the performance of search engines viz; Google, Bing and Parsijoo on Persian navigational queries Mahmoudi, Badie, Zahedi and Azimzade (2014) state that the performance of a Persian search engine "Parsijoo" is much better when Persian navigational queries are given as input in comparison to general search engines

like, Google and Bing which performs better when general English queries are given as input.

Therefore, “Parsijoo” is better search engine for those users who type their queries in a specific Persian language to attain better results. Lewandowski (2008) also evaluates the performance of five major search engines viz; Google, Yahoo, MSN, Ask.com, and Seekport while taking into consideration not only the results but also the result descriptions provided by the search engines and reveal that the performance of Google and Yahoo is comparatively betterthan other search engines.

2.2 Overview of Search Engine Optimization

(Nikhil Khokale)

The literature review will focus on exploring and analyzing the following key themes and trends related to the evolution of search engines between 2015 and 2023: Advancements in search engine algorithms: This theme will investigate the major advancements in search engine algorithms, including machine learning-based techniques such as Google's RankBrain and BERT (Bidirectional Encoder Representations from Transformers). It will analyze how these advancements have improved information retrieval, relevance, and user satisfaction. (Chen et al., 2016; Devlin et al., 2019) Personalization and user experience: This theme will explore how search engines have evolved to provide personalized search results, taking into account user preferences, context, and behaviour. It will examine features such as rich snippets, voice search, and mobile-first indexing, and their impact on enhancing user experience. (Hassan and Zaki, 2016; Zha et al., 2018) Semantic search and natural language processing: This theme will delve into the integration of semantic search techniques and natural language processing in search engine algorithms. It will assess how these technologies have enabled search engines to better understand user queries and provide more accurate and contextually relevant results. (Qiu et al., 2016; Yang et al., 2019) Voice search and virtual assistants: This theme will investigate the rise of voice search and virtual assistants, such as Google Assistant, Siri, and Alexa, and their impact on search engine design, optimization, and user interaction. It will analyze the advancements in speech recognition, natural language understanding, and voiceenabled search experiences. (Rao and Kim, 2017; Shen et al., 2020) Mobile search and local search optimization: This theme will explore the increasing importance of mobile search and the strategies employed by search engines to provide location-based and contextually relevant results. It will examine techniques such

as geolocation, mobile-friendly website optimization, and the incorporation of local business information in search results. (Li et al., 2017; Yu et al., 2019) Challenges and emerging trends: This theme will identify and analyze the challenges faced by search engines in the modern era, such as combating fake news, addressing privacy concerns, and mitigating algorithmic biases.

It will also highlight emerging trends, including the integration of artificial intelligence, augmented reality, and blockchain technology in search engine development. (Spink et al., 2016; Nguyen et al., 2021)

2.3 Web Search Engine

Search Engine:

Search Engine provides the gateway for most of the users trying to explore the huge information base of web pages. Search engines are a program that search documents for specified keywords on search for information on the World Wide Web and returns a list of the documents where the keywords were found. A Search Engine is really a class of programs; however, the term is often used to specifically describe systems like Google, Bing and Yahoo! Search that enable users to search for documents on the World Wide Web.

Goals of Search Engine:

Quality - Means effectiveness can be defined as to retrieve the most relevant set of document for a query. Process text and store text statistics to improve relevance be used. Speed - Means efficiency may be defined as a process queries from users as fast as possible For it specialized data structure should be used. How web Based Search Engine Works? Web based search engine works by saving the information of many web pages, which they retrieve itself. These pages are retrieved by a web crawler which is also called spider which follows every link on the site. Search engine is a term used for information retrieval. Search engine match queries against an index that they create. This index contains the word in each document, pointers to their location within the document.

Web search engines get their information by web crawling from site to site. The "spider" checks for the standard filename robots.txt, addressed to it, before sending certain information back to be indexed depending on many factors, such as

the titles, page content, JavaScript, Cascading Style Sheets (CSS), headings, as evidenced by the standard HTML markup of the informational content, or its metadata in HTML meta tags. Indexing means associating words and other definable tokens found on web pages to their domain names and HTML-based fields. The associations are made in a public database, made available for web search queries.

CHAPTER-3
SYSTEM ANALYSIS

3.1 Existing System

The existing system would be traditional search engines, which suffer from indexing delays and static ranking algorithms. One of the existing systems is “Cuil”. Cuil had raised $33 million in venture capital in 2007 and 2008. We first started covering Cuil in late 2007 when it was in stealth. It launched in July 2008 but a month later their VP Product had bailed. By December 2008 it had little traffic and since then things have only gotten stranger. A user could log into their Facebook account via Cuil, which would then search friend updates for topics, with search links. A user could also send messages to their friends through Cuil. It worked on an automated encyclopedia called Cpedia, built by algorithmically summarizing and clustering ideas on the web to create encyclopedia-like reports. Instead of displaying search results, Cuil would show Cpedia articles matching the searched terms. This was meant to reduce duplication by combining information into one document.

3.1.1 Drawbacks

Cuil received widely critical press coverage. Concerns were expressed about the website's slow response times, irrelevant or wrong search results. Danny Sullivan of Search Engine Watch questioned the validity of Cuil's claim that it had the world's largest search engine index and criticized it for focusing on size rather than relevance. However, despite reported problems with search results, Net Applications reported that for the last three days of July 2008, Cuil beat Google and Yahoo in the amount of time spent on a site after referral from a search engine.

After the initial critical press coverage, Cuil was alleged to have caused issues for some websites, owing to how the Cuil indexing robot polled certain sites (including under its pre-release name, Cuill). Many website owners reported that the Twiceler crawler repeatedly hit their site with randomly generated URLs in an attempt to find pages inaccessible by links. Others reported irrelevant images associated with their listing in Cuil's search results.

This was due to the behavior of its indexing robot, known as Twiceler during its pre-release phase (previously called Cuill). Website owners reported instances where the

Twiceler crawler repeatedly accessed their sites with randomly generated URLs, attempting to discover pages that weren't accessible through regular links. Additionally, some complained about irrelevant images appearing alongside their listings in Cuil's search results.

3.2 Proposed System

The proposed system is the Real-time Web Search Engine, offering dynamic fetching and updating of web content, prioritizing freshness and relevance, and employing dynamic ranking algorithms based on recency, user engagement metrics, and contextual relevance.

Leveraging advanced crawling and indexing techniques, the engine prioritizes freshness and relevance, delivering timely results that reflect the ever-changing nature of the web. Moreover, dynamic ranking algorithms factor in recency, user engagement metrics, and contextual relevance to provide personalized and contextually relevant search results. With features like personalized search experiences, real-time notifications for trending topics, and a responsive user interface, our Real-time Web Search Engine empowers users to stay informed and discover relevant content as it emerges, reshaping the way people interact with online information in today's fast-paced digital landscape.

Key features in our proposed system are :

- Constant Probing
- Real-time Cataloging
- Instantaneous Ranking
- Human-Computer Interaction (HCI)
- Instant Alerts

3.2.1 Advantages:

The proposed system offers several advantages:

Speed and Efficiency:

Our system retrieve results rapidly, allowing users to find what they need in a matter of seconds or less. This speed and efficiency save users valuable time.

Organized Results:

Our system organize search results in a structured manner, typically based on relevance, making it easier for users to find the most relevant information quickly. Our

system offers customization options, such as filters, advanced search parameters, and personalized recommendations, allowing users to tailor their search experience to their specific needs and preferences.

Continuous Improvement:

Our system continuously update its algorithms and technology to enhance search accuracy, relevance, and user experience. This ongoing improvement ensures that users receive high-quality search results.

3.3 Modules

- Continuous Crawling
- Dynamic Indexing
- Adaptive Ranking Algorithms
- User Interface
- Real-time Notifications

Continuous Crawling:

Implement a crawling mechanism to continuously fetch and update web content in real-time.

Dynamic Indexing:

Develop an indexing system to efficiently store and retrieve real-time web content.

Adaptive Ranking Algorithms:

Design dynamic ranking algorithms that prioritize freshness, relevance, and user engagement metrics.

User Interface:

Create a responsive and intuitive user interface for seamless search experiences across devices.

Real-time Notifications:

Implement a notification system to alert users to trending topics, breaking news, and updates related to their search queries.

CHAPTER-4
REQUIREMENT ANALYSIS

4.1 Software Requirements

- For developing the application the following are the software requirements

  1.Python

  2.Visual studio code

- Operating System supported

  Windows 10 64 bit OS

- Debugger and Emulator

  Any browser(particularly chrome)

4.2 Functional Requirements

- Graphical User Interface with User

4.3 Hardware Requirements

- **Processor:** Intel i5 or Above
- **Ram:** Minimum 4GB
- **Secondary storage:** Minimum 40 GB

CHAPTER-5
SOFTWARE DESIGN

5.1 Data Flow Diagram

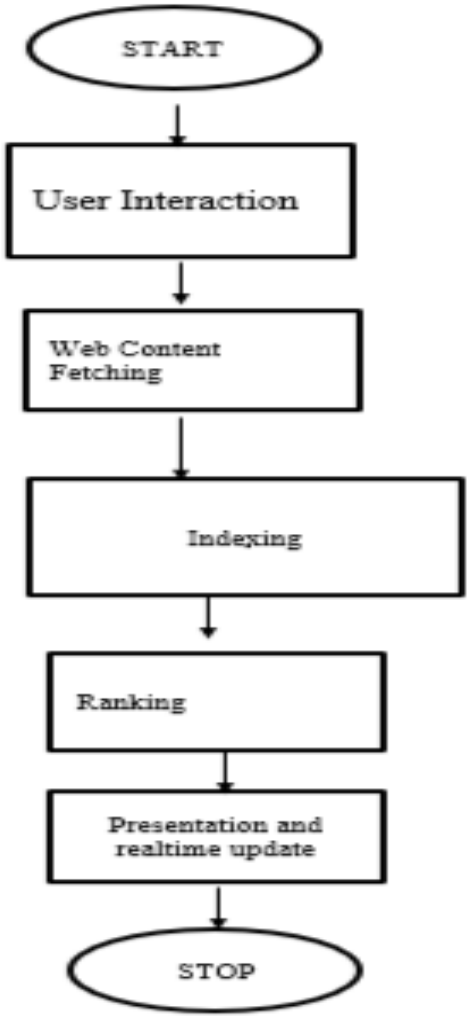


Fig 0.1 Data flow Diagram

In the context of a Real-time Web Search Engine, a DFD can be used to visualize the flow of data within the system, including how web content is fetched, indexed, ranked, and presented to users.

Identifying Processes:

Processes represent the tasks or operations performed on the data within the system. In the case of a Real-time Web Search Engine, processes may include:

Web crawling: Fetching web content from various sources in real-time.

Indexing: Storing and organizing fetched web content in the system's index.

Ranking: Applying algorithms to rank indexed web content based on relevance and other factors.

User interface: Presenting search results to users and handling user interactions.

Defining Data Stores: Data stores represent repositories where data is stored within the system. For a Real-time Web Search Engine, data stores may include:

Web content repository: Storage for fetched web content awaiting indexing.

Index: Database or data structure where indexed web content is stored.

User preferences: Database or storage for user preferences and search history.

Describing Data Flows: Data flows represent the movement of data between processes, data stores, and external entities. Examples of data flows in a Real-time Web Search Engine may include:

Web content flow: From web sources to the web content repository.

Indexed content flow: From the web content repository to the indexing process.

Search results flow: From the indexing process to the user interface for presentation to users.

Identifying External Entities: External entities represent sources or destinations of data that interact with the system. In the case of a Real-time Web Search Engine, external entities may include:

Web sources: External websites or sources from which web content is fetched.

Users: Individuals interacting with the search engine through the user interface.

5.2 Use Case Diagram

A Use Case Diagram is a visual representation that illustrates the interactions between users (actors) and a system to achieve specific goals or tasks. It depicts the various ways users interact with the system and the functionality provided by the system to fulfill those interactions. Use Case Diagrams are commonly used in software engineering to capture system requirements and to understand the system's behavior from the user's perspective.

In the context of a Real-time Web Search Engine, a Use Case Diagram can be used to identify the different types of users and their interactions with the search engine. Here's how it can be applied:

Identifying Actors: Actors represent the different types of users or external systems that interact with the Real-time Web Search Engine. Common actors in this context include:

- User: Individuals who interact with the search engine to perform searches and access search results.
- Admin: System administrators who manage and maintain the search engine, such as configuring indexing settings or monitoring system performance.

Defining Use Cases: Use cases represent specific tasks or goals that users want to accomplish using the Real-time Web Search Engine. Examples of use cases for the search engine may include:

- Perform Search: Users enter search queries and retrieve relevant search results.
- Filter Search Results: Users apply filters to refine search results based on criteria such as date, relevance, or source.
- Manage Indexing Settings: Admins configure indexing settings, such as the frequency of web crawling or the sources to include in the index.

Describing Relationships:

Relationships between actors and use cases depict how actors interact with the system to perform certain tasks. For example:

The "User" actor interacts with the "Perform Search" and "Filter Search Results" use cases to find relevant information.

The "Admin" actor interacts with the "Manage Indexing Settings" use case to configure system settings related to indexing.

User: Represents individuals who interact with the search engine to perform searches and access search results.

Admin: Represents system administrators who manage and maintain the search engine, such as configuring indexing settings or monitoring system performance.

External System: Represents other systems or sources that interact with the search engine, such as content providers or data sources.

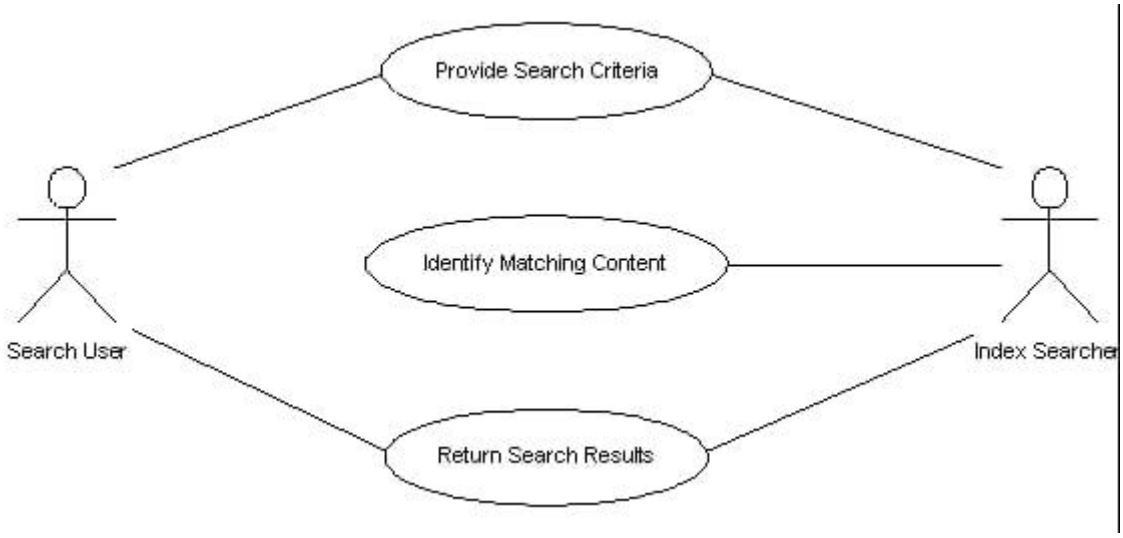


Fig 0.2 Use case diagram

Use Cases:

- Perform Search: Users enter search queries and retrieve relevant search results.
- Filter Search Results: Users apply filters to refine search results based on criteria such as date, relevance, or source.
- Manage Indexing Settings: Admins configure indexing settings, such as the frequency of web crawling or the sources to include in the index.
- Monitor System Performance: Admins monitor the performance and health of the search engine, including metrics such as response time, query throughput, and resource utilization.

This Use Case Diagram provides a high-level overview of the interactions between users, administrators, and the Real-time Web Search Engine, as well as the key functionalities supported by the system.

5.3 Class Diagram :

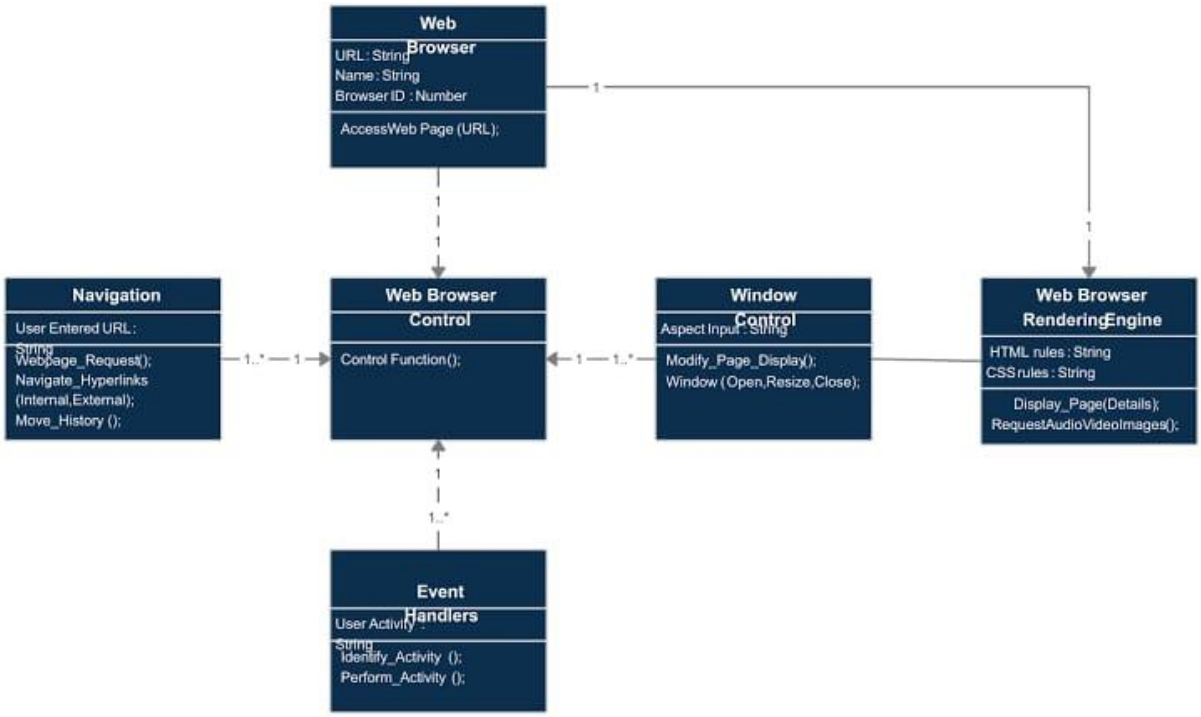


Fig 0.3 Class Diagram

5.4 Sequence Diagram :

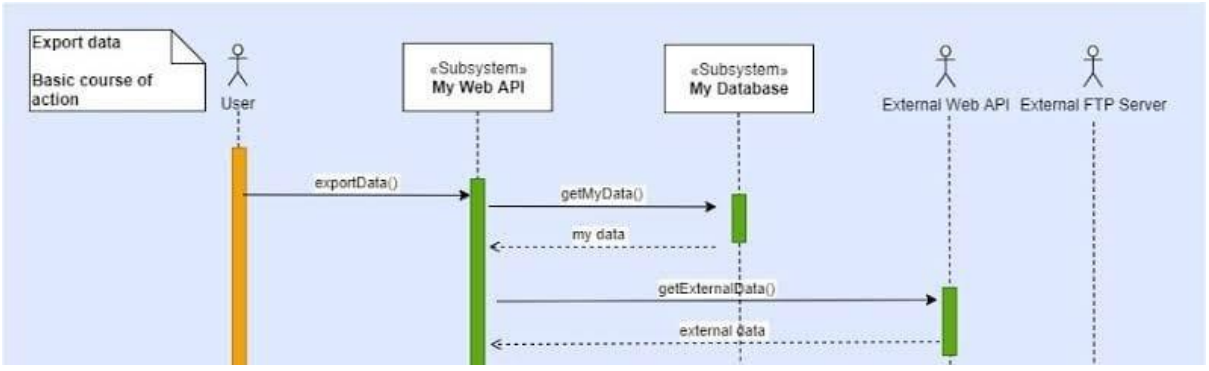


Fig 0.4 Sequence Diagram

CHAPTER-6
IMPLEMENTATION

6.1 Methodology

Algorithm 1: Real-time Crawling Algorithm

1: Begin
2: Initialize Variables: Set L to 1
3: Set Number of Feature Sentences: Determine n
4: for i = 1 to n do
5: Obtain Feature_Sentence: Feature
6: Tokenize Feature_Sentence: Tokenized_Feature
7: for Word in Tokenized_Feature do
8: Check Word Presence in Text: if Word in Text then
9: Set Feature Vector: M[L] = 1
10: else
11: Set Feature Vector: M[L] = 0
12: end if
13: Increment Index: L = L + 1
14: end for
15: end for
16: Return Feature Vector: M
17: End

Here's a step-by-step explanation of Algorithm 1,

Begin: The algorithm starts here.
Initialize Variables (Step 2-3): L is initialized to 1, and the total number of feature sentences (n) is determined.
Loop Through Feature Sentences (Step 4-15): Iterate through each feature sentence from 1 to n.
Obtain Feature Sentence (Step 5): Get the current feature sentence to process.
Tokenize Feature Sentence (Step 6): Break down the feature sentence into individual words.

Loop Through Tokenized Words (Step 7-13): Iterate through each word in the tokenized feature sentence.

Check Word Presence in Text (Step 8): Determine if the word exists in the original text.

Set Feature Vector (Step 9-11): If the word is present, set the corresponding feature value in the vector M to 1; otherwise, set it to 0.

Increment Index (Step 13): Move to the next feature index.

Return Feature Vector (Step 16): The algorithm returns the feature vector M, representing the presence or absence of each word from the feature sentences in the original text data.

End: The algorithm ends here.

Algorithm 2: Dynamic Indexing Algorithm

1: Begin
2: Initialize Variables: Set L to 1
3: Set Number of Feature Sentences: Determine n
4: for i = 1 to n do
5: Obtain Feature_Sentence: Feature
6: Tokenize Feature_Sentence: Tokenized_Feature
7: for Word in Tokenized_Feature do
8: Check Word Presence in Text: if Word in Text then
9: Set Feature Vector: M[L] = 1
10: else
11: Set Feature Vector: M[] = 0
12: end if
13: Increment Index: L = L + 1
14: end for
15: end for
16: Return Feature Vector: M
17: End

Here's a step-by-step explanation of Algorithm 2,

Begin: The algorithm starts here.

Initialize Variables (Step 2-3): L is initialized to 1, and the total number of feature sentences (n) is determined.

Loop Through Feature Sentences (Step 4-15): Iterate through each feature sentence from 1 to n.

Obtain Feature Sentence (Step 5): Get the current feature sentence to process.

Tokenize Feature Sentence (Step 6): Break down the feature sentence into individual words.

Loop Through Tokenized Words (Step 7-13): Iterate through each word in the tokenized feature sentence.

Check Word Presence in Text (Step 8): Determine if the word exists in the original text.
Set Feature Vector (Step 9-11): If the word is present, set the corresponding feature value in the vector M to 1; otherwise, set it to 0.
Increment Index (Step 13): Move to the next feature index.
Return Feature Vector (Step 16): The algorithm returns the feature vector M, representing the presence or absence of each word from the feature sentences in the original text data.
End: The algorithm ends here.

Algorithm 3: Dynamic Ranking Algorithm

1: Begin
2: Initialize Variables: Set L to 1
3: Set Number of Feature Sentences: Determine n
4: for i = 1 to n do
5: Obtain Feature_Sentence: Feature
6: Tokenize Feature_Sentence: Tokenized_Feature
7: for Word in Tokenized_Feature do
8: Check Word Presence in Text: if Word in Text then
9: Set Feature Vector: M[L] = 1
10: else
11: Set Feature Vector: M[L] = 0
12: end if
13: Increment Index: L = L + 1
14: end for
15: end for
16: Return Feature Vector: M
17: End

Here's a step-by-step explanation of Algorithm 3,

Begin: The algorithm starts here.

Initialize Variables (Step 2-3): L is initialized to 1, and the total number of feature sentences (n) is determined.

Loop Through Feature Sentences (Step 4-15): Iterate through each feature sentence from 1 to n.

Obtain Feature Sentence (Step 5): Get the current feature sentence to process.

Tokenize Feature Sentence (Step 6): Break down the feature sentence into individual words.

Loop Through Tokenized Words (Step 7-13): Iterate through each word in the tokenized feature sentence.

Check Word Presence in Text (Step 8): Determine if the word exists in the original text.

Set Feature Vector (Step 9-11): If the word is present, set the corresponding feature value in the vector M to 1; otherwise, set it to 0.

Increment Index (Step 13): Move to the next feature index.

Return Feature Vector (Step 16): The algorithm returns the feature vector M, representing the presence or absence of each word from the feature sentences in the original text data.

End: The algorithm ends here.

6.2 Source Code

```
import streamlit as st
import sqlite3
from sentence_transformers import SentenceTransformer
from sklearn.metrics.pairwise import cosine_similarity
import json

# Connect to SQLite database
conn = sqlite3.connect('data.db')
cursor = conn.cursor()

# Create table if not exists
cursor.execute('''
    CREATE TABLE IF NOT EXISTS sentences (
        id INTEGER PRIMARY KEY AUTOINCREMENT,
```

```
        url TEXT,
        description TEXT,
        embedding TEXT
    )
''')
conn.commit()

# Load Sentence Transformer model
model = SentenceTransformer('paraphrase-MiniLM-L6-v2')



# Streamlit App
def add_data_page():
    st.title('Add Data to Database')

    url = st.text_input('Enter URL:')
    description = st.text_area('Enter Description:')

    if st.button('Add to Database'):
        # Generate sentence embedding
        embedding = model.encode(description, convert_to_tensor=True).tolist()

        # Convert the embedding list to a JSON string
        embedding_json = json.dumps(embedding)

        # Save data to SQLite database
        cursor.execute('INSERT INTO sentences (url, description, embedding) VALUES (?,
?, ?)',
                       (url, description, embedding_json))
        conn.commit()
```

```
        st.success('Data added to the database!')

def query_data_page():
    st.title('Query Database')

    query = st.text_input('Enter query:')

    if st.button('Run Query'):
        # Generate sentence embedding for the query
        query_embedding = model.encode(query, convert_to_tensor=True).tolist()

        # Fetch data from the database and calculate similarity
        cursor.execute('SELECT id, url, description, embedding FROM sentences')
        results = cursor.fetchall()
        if results:
            similarities = []
            for result in results:
                # Deserialize the embedding from JSON
                stored_embedding = json.loads(result[3])

                # Calculate cosine similarity
                similarity = cosine_similarity([query_embedding], [stored_embedding])[0][0]
                similarities.append((result, similarity))

            # Sort results by similarity
            similarities.sort(key=lambda x: x[1], reverse=True)

            # Display results
            st.write('Results:')
            for result, similarity in similarities:
```

```
            st.write(f'ID: {result[0]}, URL: {result[1]}, Description: {result[2]}, Similarity: {similarity:.4f}')
        else:
            st.warning('No data in the database.')


# Streamlit main app
def main():
    st.sidebar.title('Search Engine')
    page = st.sidebar.radio('Select a page:', ('Add Data', 'Query Data'))
    if page == 'Add Data':
        add_data_page()
    elif page == 'Query Data':
        query_data_page()
if _name_ == '_main_':
    main()
```

CHAPTER-7
TESTING

7.1 System Test

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub-assemblies, assemblies and/or a finished product It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

7.2 Types Of Tests

7.2.1 Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application .it is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

7.2.2 Integrated Testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfaction, as shown by successfully unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

7.2.3 Functional test

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

1. **Valid Input:** Identified classes of valid input must be accepted.
2. **Invalid Input:** Identified classes of invalid input must be rejected
3. **Functions:** Identified functions must be exercised.
4. **Output:** Identified classes of application outputs must be exercised.
5. **Systems/Procedures:** Interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

7.2.4 White Box Testing

White Box Testing is a testing in which in which the software tester has knowledge of the inner workings, structure, and language of the software, or at least its purpose. It is purpose. It is used to test areas that cannot be reached from a black box level.

7.2.5 Black Box Testing

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box. you cannot “see” into it. The test provides inputs and responds to outputs without considering how the software works.

7.3 Test strategy and approach

Field testing will be performed manually and functional tests will be written in detail.

7.3.1 Test objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

7.4 Acceptance Testing

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

7.3.2 Features to be tested

- Verify that the entries are of the correct format
- No duplicate entries should be allowed
- All links should take the user to the correct page.

7.5 Test Results:

All the test cases mentioned above passed successfully. No defects encountered.

OUTPUT

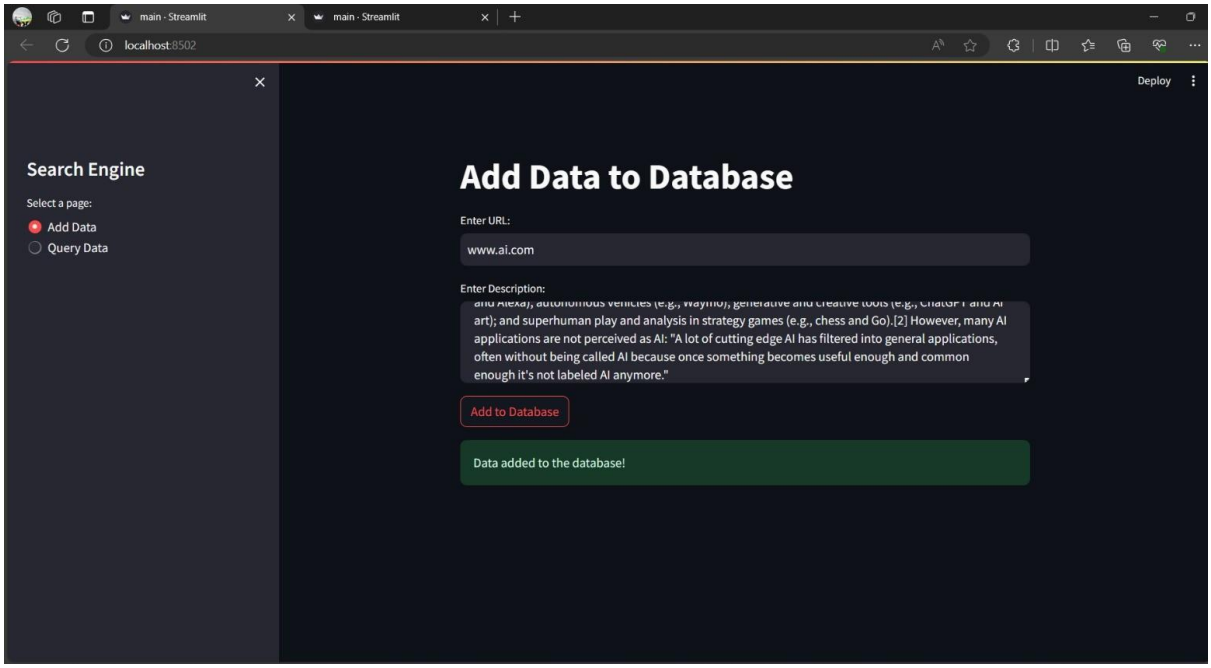


Fig 7.1 Positive Output

CHAPTER-8
RESULTS

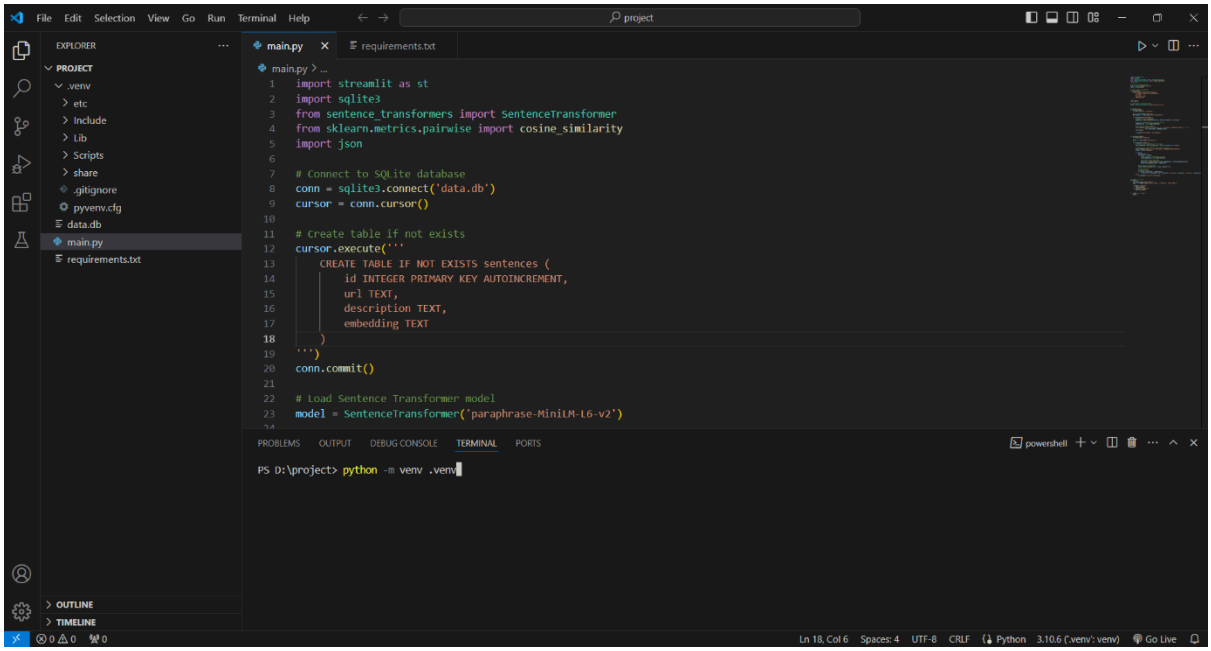


Fig 0.1 Create an virtual environment By using the syntax: python –m venv myenv

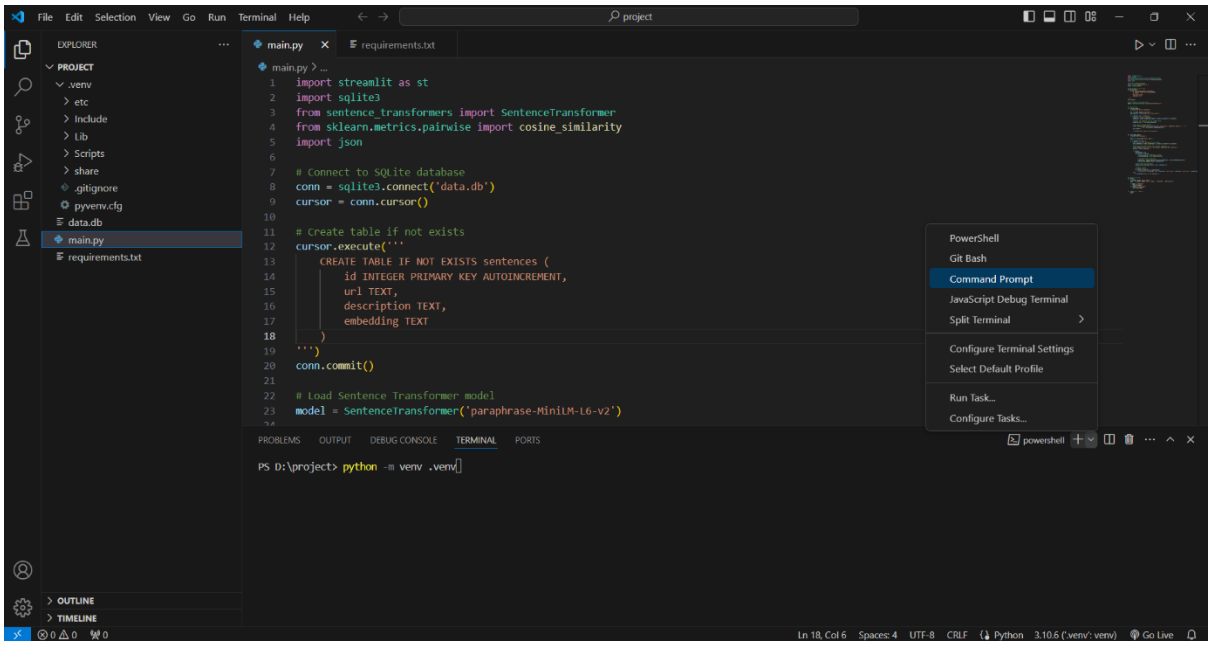


Fig 0.2 Open Command Prompt to activate venv

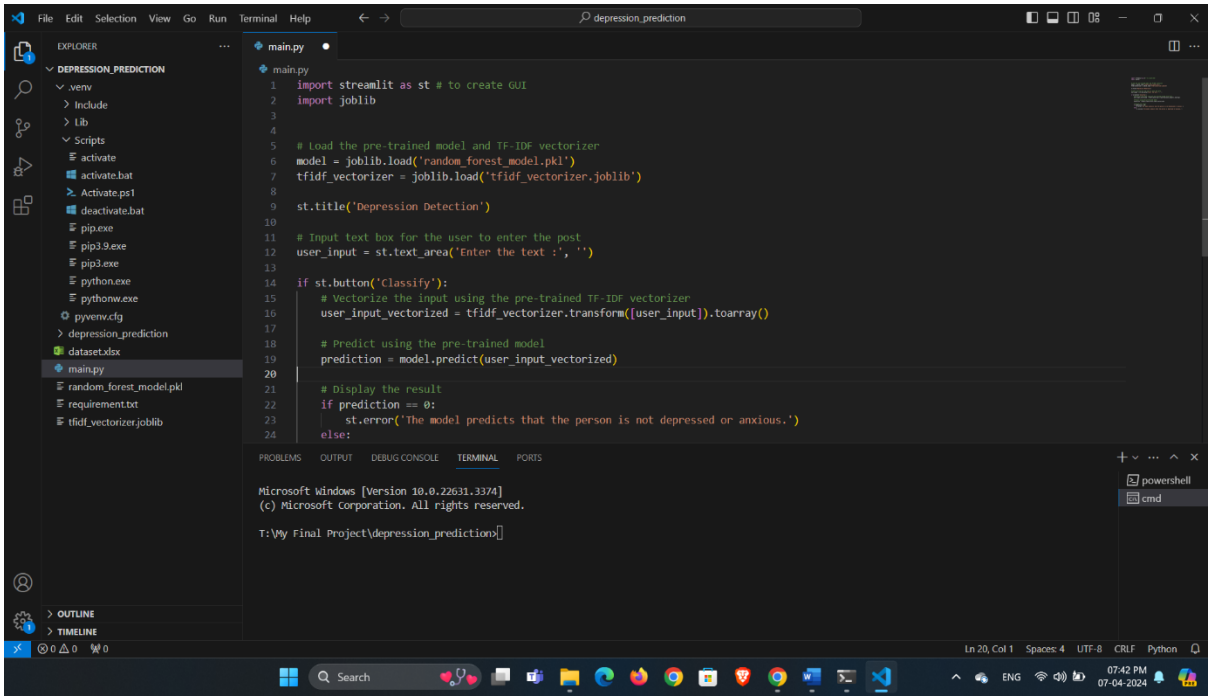


Fig 0.3 Acivate venv

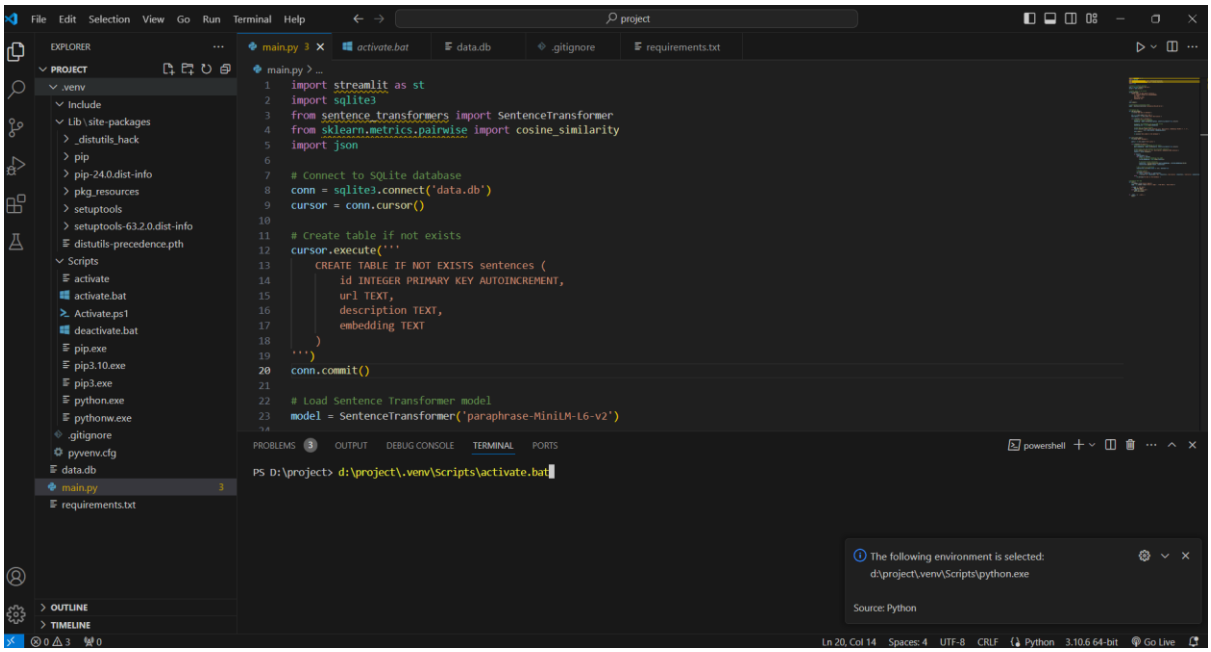


Fig 0.4 To pip install requirements by using the syntax: pip install –r requirements.txt

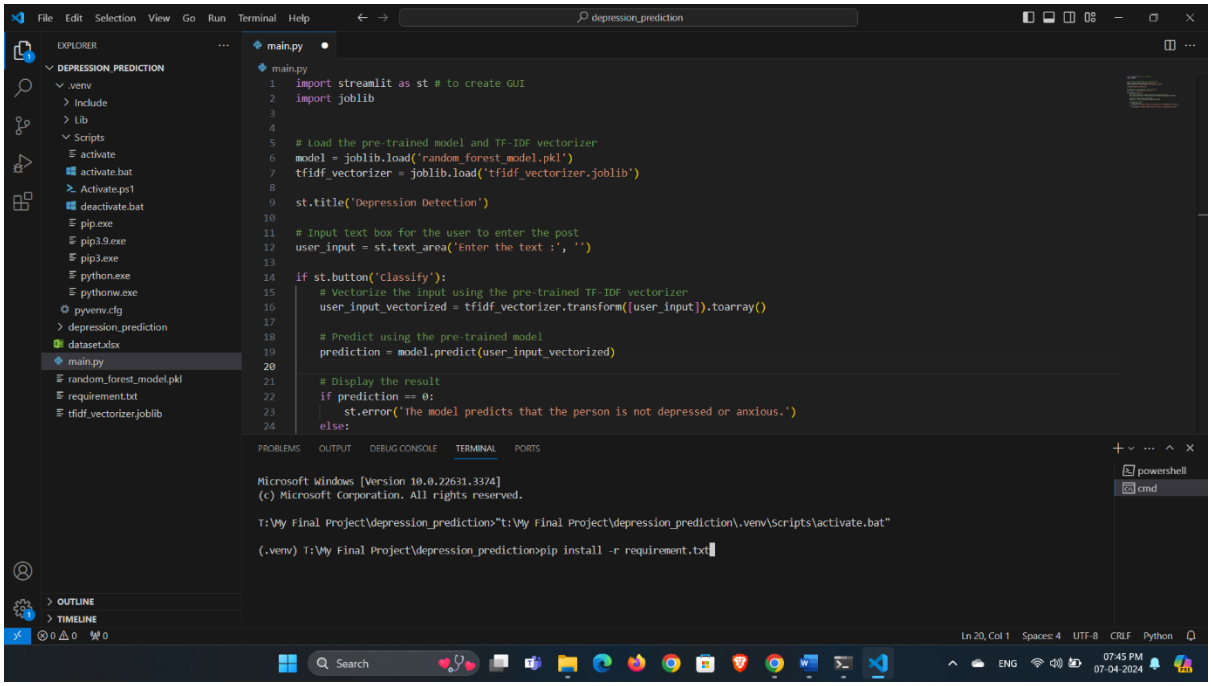


Fig 0.5 Press Enter

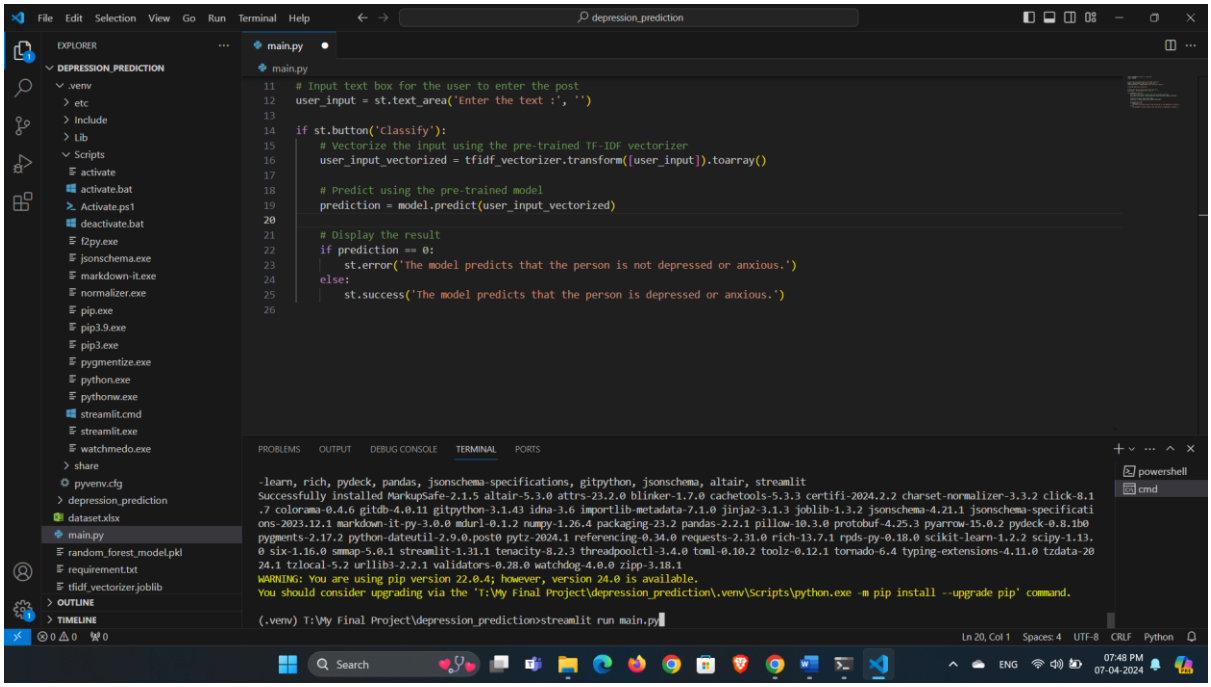


Fig 0.6 Collecting Requirements

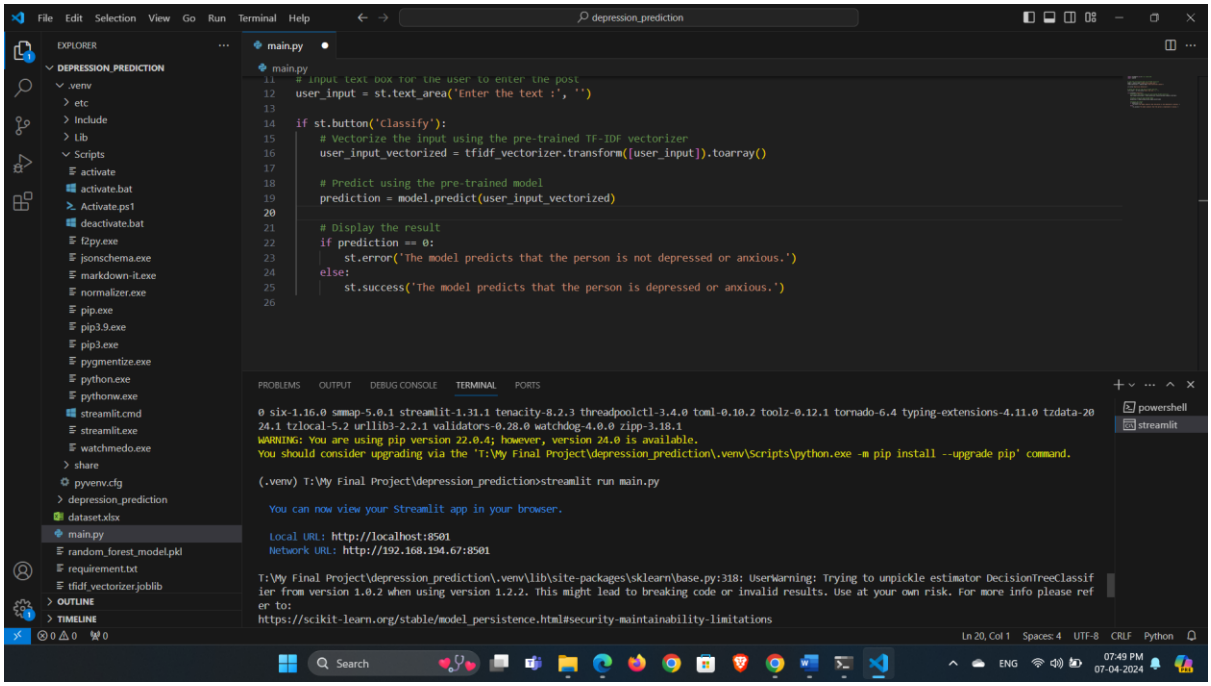


Fig 0.7 Run the main file streamlit run main.py

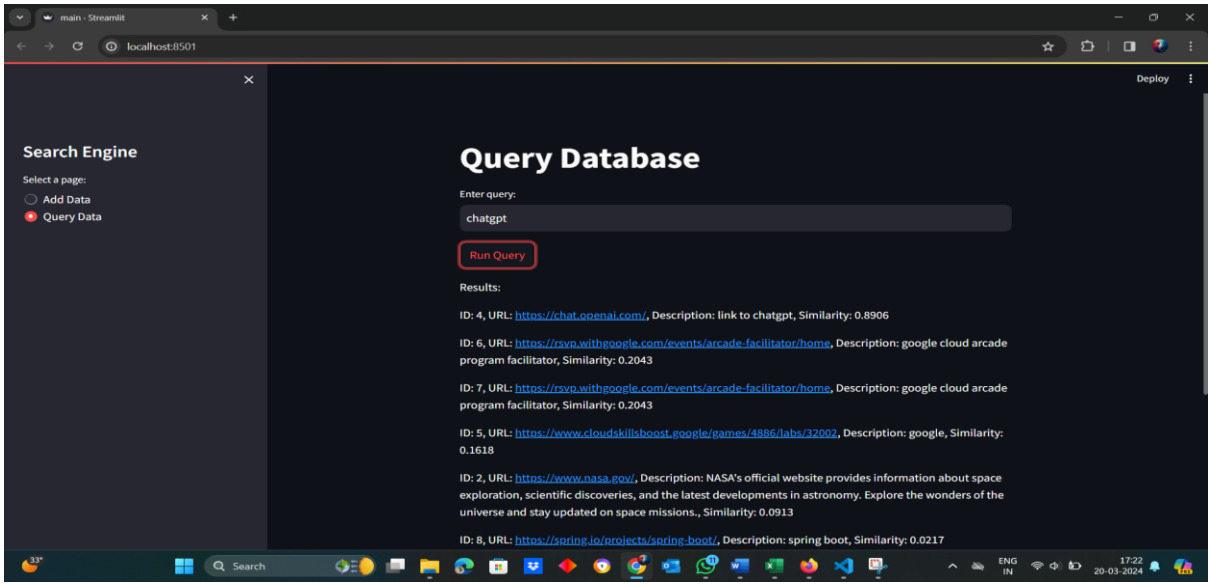


Fig 0.8 Then obtained the final output web page

CHAPTER-9
CONCLUSION & FUTURE ENHANCEMENT

The real-time web search engine represents a pivotal advancement in the realm of information retrieval, offering users unparalleled access to up-to-date and relevant web content. Through continuous crawling, indexing, and dynamic ranking algorithms, this innovative platform has redefined the way users interact with online information, enabling them to stay informed, connected, and engaged in today's fast-paced digital landscape.

By empowering users to perform searches, filter search results, and access personalized content in real-time, the search engine has revolutionized the process of discovering and consuming information on the web. Moreover, the system's robust architecture, coupled with advanced monitoring and management capabilities, ensures optimal performance, scalability, and reliability, even under the most demanding conditions.

In conclusion, the real-time web search engine stands as a testament to the power of technology in democratizing access to information and empowering individuals in their quest for knowledge. With its real-time capabilities and user-centric design, the search engine not only facilitates information discovery but also fosters greater connectivity and collaboration in the digital age, paving the way for A more informed and enlightened society.

Advanced machine learning integration:

Incorporate machine learning algorithms to enhance search relevance, personalization, and user engagement. Algorithms could learn from user behavior, preferences, and context to continuously improve search results.

Semantic search capabilities:

Integrate semantic search techniques to understand the meaning and context of search queries, enabling the search engine to provide more accurate and contextually relevant results.

Voice search and natural language processing (nlp):

Implement voice search functionality and NLP capabilities to allow users to perform searches using natural language queries, improving accessibility and user experience.

Real-time trend analysis and prediction:

Develop capabilities to analyze real-time web trends and predict emerging topics or events,

enabling the search engine to provide proactive recommendations and alerts to users.

Enhanced content aggregation and summarization:

Expand content aggregation capabilities to include a wider range of sources and formats (e.G., Social media, multimedia content). Additionally, develop content summarization algorithms to provide concise summaries of search results.

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